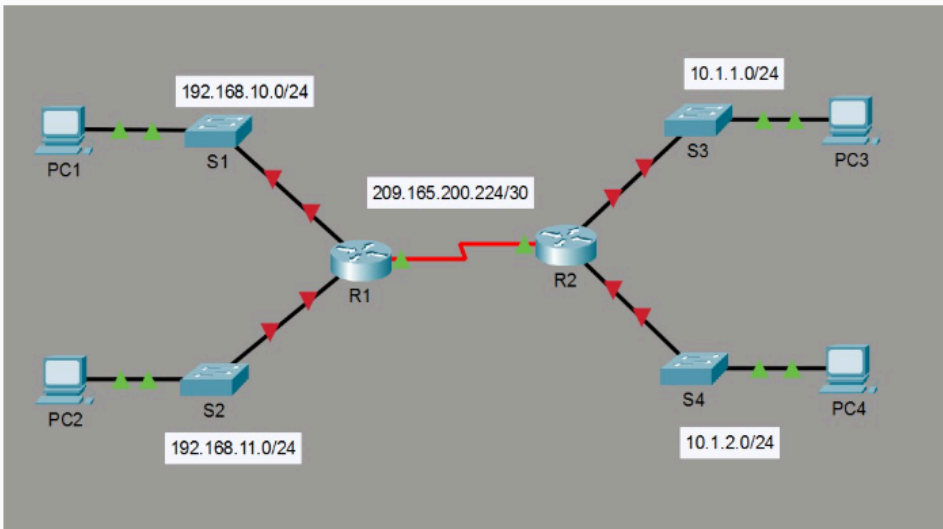


11.0 Summative Assessment: Packet Tracer Lab 8: Connect a Router to a LAN

In this lab, I configured multiple router interfaces across several LANs and a WAN connection. I used verification commands to inspect interfaces, review routing tables, and test end-to-end connectivity across the network.

This lab helped reinforce how routers connect multiple networks and how routing tables help determine packet paths.

Network Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.10.1	255.255.255.0	N/A
	G0/1	192.168.11.1	255.255.255.0	N/A
	S0/0/0 (DCE)	209.165.200.225	255.255.255.252	N/A
R2	G0/0	10.1.1.1	255.255.255.0	N/A
	G0/1	10.1.2.1	255.255.255.0	N/A
	S0/0/0	209.165.200.226	255.255.255.252	N/A
PC1	NIC	192.168.10.10	255.255.255.0	192.168.10.1
PC2	NIC	192.168.11.10	255.255.255.0	192.168.11.1
PC3	NIC	10.1.1.10	255.255.255.0	10.1.1.1
PC4	NIC	10.1.2.10	255.255.255.0	10.1.2.1

Part 1: Display Router Information

Access R1:

```
Router> enable  
Password: class  
Router#
```

View All Interface Statistics

```
R1# show interfaces
```

Displays all router interfaces.

View Serial Interface Information

```
R1# show interfaces serial 0/0/0
```

Use this to verify:

- IP address
- bandwidth
- interface status

View Gigabit Interface Information

```
R1# show interfaces gigabitEthernet 0/0
```

Use this to verify:

- IP address
- MAC address
- bandwidth

View Interface Summary

```
R1# show ip interface brief
```

This displays:

- interface names
- IP addresses
- status
- protocol status

View Routing Table

```
R1# show ip route
```

Look for:

```
C
```

Connected routes

```
0
```

OSPF learned routes

Part 2: Configure Router Interfaces

Configure R1 G0/0

```
R1# configure terminal
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# description LAN connection to S1
R1(config-if)# exit
```

Configure R1 G0/1

```
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip address 192.168.11.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# description LAN connection to PC2 network
R1(config-if)# exit
```

Configure R2 G0/0

```
R2# configure terminal
R2(config)# interface gigabitEthernet 0/0
R2(config-if)# ip address 10.1.1.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# description LAN connection to PC3 network
R2(config-if)# exit
```

Configure R2 G0/1

```
R2(config)# interface gigabitEthernet 0/1
R2(config-if)# ip address 10.1.2.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# description LAN connection to PC4 network
R2(config-if)# exit
```

Save Configurations

R1:

```
R1# copy running-config startup-config
```

R2:

```
R2# copy running-config startup-config
```

Part 3: Verify Configuration

Verify Interface Status

```
R1# show ip interface brief
R2# show ip interface brief
```

Confirm interfaces show:

```
up/up
```

Verify Routing Tables

```
R1# show ip route
R2# show ip route
```

Verify:

- Connected routes
- OSPF routes
- WAN routes
- LAN routes

Test Connectivity

From PC1:

```
C:\> ping 10.1.2.10
```

From R2:

```
R2# ping 192.168.11.10
```

All devices should successfully communicate.

Troubleshooting

If interfaces show:

```
administratively down
```

Fix:

```
R1(config-if)# no shutdown
```

If pings fail:

Check:

- IP addresses
- subnet masks
- default gateways
- interface status
- routing table entries

If routes are missing:

```
R1# show ip route
```

Verify OSPF learned routes exist.

Commands Used (In Order)

```
enable
show interfaces
show interfaces serial 0/0/0
show interfaces gigabitEthernet 0/0
show ip interface brief
show ip route

configure terminal

interface gigabitEthernet 0/0
ip address
no shutdown
description
exit

interface gigabitEthernet 0/1
ip address
no shutdown
description
exit

copy running-config startup-config

ping
```

Key Takeaways

This lab reinforced:

- LAN routing
- WAN connectivity
- interface configuration
- routing table analysis
- OSPF verification
- end-to-end troubleshooting

This felt like one of my first true multi-network routing labs where everything began working together.

<https://herbal-rancher.github.io/portfolio/walkthroughs/lesson-2-0/lab-8-connect-router-to-lan/>